



International Journal of Multidisciplinary Science, Engineering and Technology

Volume 1 | Issue 1 | Sept 2026

CHIEF EDITOR
Anuj Verma



International Journal of Multidisciplinary Science, Engineering and Technology

IJMSET | International Peer-Reviewed Open Access Journal

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Chief Editor Message

It gives me great pleasure to welcome you to the inaugural issue of the International Journal of Multidisciplinary Science, Engineering & Technology (IJMSET), Volume 1, Issue 1, Sept 2026.

IJMSET has been established as a platform for researchers, academicians, scientists, engineers and professionals to share original research, innovative ideas and emerging developments in science, engineering, and technology.

We are committed to maintaining high standards of research quality, originality, academic integrity and peer review. Through the journal, we aim to encourage interdisciplinary research and promote the exchange of knowledge that contributes to scientific and technological advancement.

I sincerely thank the authors, reviewers, editorial board members and well-wishers for your valuable support in making this inaugural issue possible.

I warmly invite researchers and professionals to contribute your quality research to future issues of IJMSET.

We look forward to building IJMSET into a meaningful platform for scholarly communication and innovation.

Chief Editor
IJMSET

International Journal of Multidisciplinary Science, Engineering and Technology
(IJMSET)

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Development of Ultra-High-Performance Concrete (UHPC): Materials Science, Engineering Practice and Sustainability Challenges

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Abstract

Ultra-high-performance concrete (UHPC) is a class of fiber-reinforced cementitious composites characterized by compressive strengths above 150 MPa, tensile strength and post-cracking ductility supplied by steel fibers, and extremely low permeability. This paper reviews the development of UHPC from its origins in densified cement systems of the 1970s and 1980s to its present use in bridges, architecture, and repair. It examines the design principles behind the material (particle packing, low water-to-binder ratio, elimination of coarse aggregate, fiber reinforcement, and curing), summarizes typical mechanical and durability properties, and surveys applications and standardization. The principal barriers to wider adoption, namely cost, embodied carbon, shrinkage, and quality control, are discussed alongside current research on low-carbon binders, ambient curing, 3D printing, and multifunctional composites. UHPC has matured from a laboratory material into an engineering product, and the central challenge is now delivering it economically and sustainably.

Keywords: ultra-high-performance concrete; UHPFRC; reactive powder concrete; particle packing; steel fibers; durability; low-carbon concrete

1. Introduction

Conventional concrete is the most widely used construction material in the world, yet its typical compressive strength (20-50 MPa), low tensile capacity, brittle failure, and moderate durability limit slender design and long service life. Its heterogeneous microstructure, containing coarse aggregate, a weak interfacial transition zone (ITZ), capillary porosity, and microcracks, is the underlying cause of these limits.

UHPC addresses these limits by redesigning the material at the particle scale. Coarse aggregate is removed, the granular skeleton is optimized for density, water content is cut sharply with the help of superplasticizers, and short steel fibers are added to control cracking. The result is a material with compressive strength commonly between 150 and 250 MPa, flexural strength of 20-50 MPa, and durability properties far beyond those of high-performance concrete.

This paper traces how UHPC developed, explains the science behind its performance, and assesses where the field is heading. Section 2 covers historical development; Section 3 material design principles; Section 4 properties; Section 5 applications and standards; Section 6 limitations; Section 7 current research directions; and Section 8 concludes.

2. Historical Development

2.1 Foundations (1970s-1980s)

The ideas behind UHPC grew from three advances: the arrival of high-range water-reducing admixtures (superplasticizers) in the 1970s, the use of silica fume as a highly reactive ultrafine filler, and recognition that strength depends strongly on porosity. Work in Denmark on “densified systems containing homogeneously arranged ultrafine particles” (DSP) demonstrated that cement pastes with very low water content and packed ultrafine particles could reach strengths far above those of ordinary concrete (Bache, 1981). Macro-defect-free (MDF) cements, in which polymers and processing removed large flaws, pointed to the same conclusion: eliminating defects is as important as adding strength.

2.2 Reactive Powder Concrete (1990s)

The first material widely regarded as UHPC was reactive powder concrete (RPC), developed at Bouygues in France (Richard and Cheyrezy, 1995). Its design principles were:

1. removal of coarse aggregate, with maximum grain size of roughly 0.6 mm or less, to improve homogeneity;
2. optimization of the granular mixture for maximum packing density;
3. use of silica fume for pozzolanic reaction and void filling;
4. very low water-to-binder ratio (about 0.15-0.20);
5. heat treatment (and, in laboratory versions, pressure applied during setting) to accelerate reactions and densify the matrix; and
6. addition of small steel fibers for ductility.

Laboratory RPC reached compressive strengths of about 200 MPa, and with pressure and high-temperature curing and metallic aggregates, reported values approached 800 MPa. The Sherbrooke pedestrian bridge in Quebec (1997), a post-tensioned open-web space truss built from RPC, was a landmark first application (Blais and Couture, 1999).

2.3 Commercialization and Standardization (2000s)

Proprietary products followed, including Ductal (Lafarge, Bouygues, Rhodia), CEMTEC and BSI in France, and Densit-based materials in Denmark. France issued the first national recommendations for UHPC (AFGC/SETRA, 2002, revised in 2013). A large German priority program centered at the University of Kassel advanced the use of ultrafine particles, nano-silica, and improved fiber-matrix bond, and summarized its results in a comprehensive monograph (Fehling et al., 2014). Japan and South Korea built early structures such as the Sakata-Mirai footbridge and the Seonyu Footbridge in Seoul (2002).

2.4 Broader Adoption (2010s to present)

In the United States, the Federal Highway Administration (FHWA) systematically characterized UHPC and promoted it, particularly for accelerated bridge construction (Graybeal, 2006; Russell and Graybeal, 2013). Field-cast UHPC joints between precast bridge

deck panels have become the most common use because the material's bond and tensile capacity permit short lap splices and thin, durable connections. Test standards such as ASTM C1856 (fabrication and testing of specimens) and ASTM C1857 (flow measurement) were developed, and Switzerland (SIA 2052), Australia, and other countries have published guidance. Meanwhile, researchers showed that UHPC-level strength could be achieved with conventional materials and ambient or modest curing (Wille et al., 2011), and "non-proprietary" mixes emerged to lower cost.

3. Material Design Principles

3.1 Particle packing

The central principle is maximizing solid-particle density so that little space remains for water-filled pores. Continuous grading models (Fuller and Thompson, 1907; Andreasen and Andersen, 1930), their modified forms with a minimum particle size (Funk and Dinger, 1994), and compressible packing models (de Larrard and Sedran, 1994) are used to proportion cement (about 10-100 μm), quartz or limestone powder, silica fume (about 0.1-1 μm), and fine sand (typically below 1-2 mm). Because gaps between larger grains are filled by finer particles, less paste is needed to fill voids, and the water demand for flow falls.

3.2 Low water-to-binder ratio and superplasticizers

Water-to-binder ratios are typically 0.14-0.22. At these values there is insufficient water to hydrate all cement, so a portion of the clinker acts as a strong, unhydrated filler. Polycarboxylate-ether superplasticizers disperse the fine particles and provide self-consolidating flow despite the low water content. This flow supports casting into thin sections and around dense reinforcement.

3.3 Elimination of coarse aggregate

Coarse aggregate concentrates stress, produces weak ITZs, and generates microcracks from differential stiffness and thermal or shrinkage mismatch. Removing it yields a more homogeneous matrix with higher tensile strength and better resistance to crack initiation.

3.4 Supplementary cementitious materials

Silica fume contributes pozzolanic reaction (consuming portlandite to form additional calcium silicate hydrate) and a physical filler effect. Ground quartz acts as a filler and, under heat curing, participates in reaction. Fly ash, ground granulated blast-furnace slag, limestone powder, and calcined clays can replace part of the cement, and nano-silica or nano- CaCO_3 can further refine the microstructure.

3.5 Fiber reinforcement

A matrix this strong is inherently brittle, so fibers are essential to structural use. Steel fibers of about 13 mm length and 0.2 mm diameter, at roughly 2% by volume, are typical. They bridge

cracks, transfer stress, and give substantial post-cracking capacity, with strain-hardening behavior in tension in well-designed mixes and fracture energies of roughly 20-40 kJ/m², two orders of magnitude greater than plain concrete. Fiber dispersion and orientation strongly influence tensile performance and depend on casting method and flow, so this is a design variable rather than a constant.

3.6 Curing

Thermal curing at about 90 °C for one to two days accelerates the pozzolanic reaction and can raise strength and reduce long-term shrinkage. Because it demands energy and controlled facilities, ambient-cured formulations have been pursued, particularly for cast-in-place work, though they generally develop strength more slowly.

4. Properties

4.1 Mechanical behavior

Property	Typical UHPC	Conventional concrete
Compressive strength	150-250 MPa	20-50 MPa
Flexural strength	20-50 MPa	3-6 MPa
Direct tensile strength (post-cracking)	8-15 MPa	2-4 MPa
Elastic modulus	45-60 GPa	25-35 GPa
Fracture energy	20-40 kJ/m ²	~0.1 kJ/m ²

The compressive stress-strain response is nearly linear to peak, after which failure can be sudden unless fibers or confinement provide ductility. Under tension, fiber-reinforced UHPC shows an elastic branch, then multiple fine cracking and a hardening or softening phase governed by fiber pullout.

4.2 Durability

The dense, discontinuous pore structure gives UHPC chloride diffusion coefficients and water permeability that are orders of magnitude lower than conventional concrete. It shows strong resistance to freeze-thaw cycling, deicer scaling, carbonation, sulfate attack, alkali-silica reaction, and abrasion. Small cracks that do form are fine and tend to self-heal partly through continued hydration of unreacted clinker. These properties underpin claims of service lives well beyond those of conventional structures and reduced maintenance.

4.3 Time-dependent behavior

Autogenous shrinkage is large because of the very low w/b ratio and self-desiccation, commonly several hundred microstrain within the first days. Creep is lower than for conventional concrete once thermal curing has been applied. Shrinkage-reducing admixtures, expansive agents, internal curing, and thermal treatment are used to manage cracking risk.

5. Applications and Standards

Bridges. Field-cast joints between precast deck elements, thin deck overlays, precast girders and piles, and complete bridges (for example, the Mars Hill Bridge in Wapello County, Iowa, and numerous structures in France, Japan, South Korea, and Canada).

Architecture. Slender, complex-form elements such as the facade lattice of the MuCEM in Marseille and the dome components of the Louvre Abu Dhabi, where strength allows thin sections and where surface quality permits fine finishes.

Repair and strengthening. Thin overlays and jackets on existing bridges, piers, and marine structures, where the material's bond, impermeability, and abrasion resistance extend life.

Protective structures. Blast- and impact-resistant panels, owing to high energy absorption.

Energy and offshore. Wind turbine tower sections and connections, marine and nuclear-related components requiring durability.

Design and testing rules remain less mature than for conventional concrete. Documents include the French AFGC recommendations (2013), the Swiss SIA 2052, ASTM C1856 and C1857, FHWA guidance, and ongoing work in ACI Committee 239 and *fib*. Because fiber orientation and tensile behavior depend on how a member is cast, standards increasingly require structure-specific test panels or member-level validation, which slows uptake.

6. Challenges and Limitations

1. **Cost.** Cement-rich mixes, silica fume, quartz powder, and steel fibers make UHPC several times more expensive per unit volume than conventional concrete. Life-cycle analyses often favor UHPC where thinner sections, longer service life, or faster construction offset the price, but first-cost constraints hinder acceptance.
2. **Embodied carbon.** Binder contents of roughly 700-1000 kg/m³ yield high carbon intensity per cubic meter. Per unit of structural performance the result can be competitive, since less material is needed, but this depends on the design.
3. **Shrinkage and cracking.** Early-age autogenous shrinkage and restrained deformation require careful mitigation.
4. **Production and quality control.** High-shear mixers, long mixing times, sensitivity to temperature and water content, and the need to control fiber dispersion and orientation demand skilled batching and placement.
5. **Incomplete codes.** Lack of harmonized design provisions and contractor experience deters owners and designers.
6. **Compressive brittleness.** Without fibers or confinement, failure in compression can be explosive.

7. **Fiber-related concerns.** Steel fibers can be a hazard to workers, complicate recycling, and, at exposed surfaces, corrode cosmetically.

7. Current Research Directions

Low-carbon UHPC. Partial replacement of cement with slag, fly ash, calcined clay and limestone (LC³-type systems), recycled glass powder, and other supplementary materials; ecologically oriented particle-packing design that reduces binder content (Yu et al., 2014); alkali-activated UHPC; and CO₂ mineralization.

Local-materials and non-proprietary mixes. Adapting regional sands, cements, and SCMs to reduce cost and supply-chain dependence, supported by performance-based specifications.

Ambient or low-energy curing. Chemical activators, finer or more reactive binders, and calcium-aluminate or sulfoaluminate blends to remove steam curing.

Advanced fibers. Hybrid steel-synthetic systems, twisted or hooked fibers with improved pullout, recycled tire-steel fibers, natural fibers, and magnetic alignment of fibers to control anisotropy.

Additive manufacturing. Printable UHPC that combines high strength with reduced formwork, and which requires adjusted rheology and fiber handling.

Multifunctional and self-sensing composites. Carbon nanotubes, graphene, carbon fibers, and conductive additions for structural health monitoring, heating, and shielding.

Shrinkage control. Superabsorbent polymers, lightweight fine aggregate for internal curing, and expansive agents.

Modeling and design. Multiscale and mesoscale simulation of fiber-matrix interaction, machine-learning optimization of mixes, probabilistic models of fiber orientation, and performance-based design frameworks.

Hybrid systems. Textile-reinforced UHPC and UHPC combined with conventional or high-strength rebar for lightweight, thin structural systems.

8. Conclusion

UHPC developed through a sequence of connected advances: superplasticizers and silica fume made dense low-water pastes possible in the 1970s and 1980s; the RPC concept of eliminating coarse aggregate and optimizing packing defined the material in the 1990s; and commercial products, national recommendations, and bridge applications established it in the 2000s and 2010s. Its microstructural efficiency yields strength, ductility, and durability that permit slender, long-lived structures with reduced maintenance.

Wider adoption depends on solving problems that are largely economic and environmental rather than scientific. The most consequential research targets are lower-carbon binders, non-

proprietary mixes built from local materials, ambient curing, and reliable methods to predict and control fiber orientation. Parallel progress in codes and in contractor training will be required to let designers specify UHPC with confidence. If these are achieved, UHPC may shift from a specialty material for critical connections and landmark structures to a mainstream option for resilient, resource-efficient infrastructure.

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Utilization of Fly Ash for the Production of Artificial Aggregates: Manufacturing Techniques, Engineering Properties and Sustainability in Concrete

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Abstract

The growing demand for construction aggregates has increased pressure on natural resources and highlighted the need for alternative materials with lower environmental impacts. Fly ash, a by-product of coal combustion in thermal power plants, has potential for use in manufacturing artificial aggregates through different processing techniques. This review examines the application of fly ash as a principal constituent in artificial aggregate production, with particular attention to cold bonding, sintering and alkali-activated processes. The effects of raw material characteristics, binder composition, pelletization conditions and curing regimes on aggregate properties are discussed. Published studies indicate that fly ash-based aggregates can achieve lower specific gravity than conventional aggregates, making them potential constituents of lightweight concrete. However, their water absorption, crushing resistance, durability and environmental performance depend on the production method and the chemical and physical characteristics of the starting materials. The review also considers the implications of incorporating these aggregates into concrete, including fresh properties, mechanical strength, thermal performance and long-term durability. Important research needs include standardization of production parameters, assessment of leaching behaviour, long-term performance evaluation and life-cycle analysis. The findings provide a framework for evaluating fly ash as a resource for engineered aggregates and identifying areas for further experimental investigation.

Keywords: Fly ash; artificial aggregates; cold bonding; sintering; lightweight concrete; industrial waste; sustainable construction; aggregate performance.

1. Introduction

Aggregates form a major proportion of concrete and play an important role in determining its physical, mechanical and durability characteristics. The continued expansion of infrastructure has increased the demand for coarse and fine aggregates, creating concerns about the depletion of natural resources and the environmental impacts of extraction. The development of artificial aggregates from industrial by-products is one approach to reducing dependence on conventional aggregate sources while creating opportunities for waste utilization.

Fly ash is a finely divided material collected from the flue gases of coal-fired power plants. Its composition and reactivity depend on the source of coal, combustion conditions and ash collection system. The presence of aluminosilicate phases and, in some cases, calcium-bearing compounds makes fly ash a potential feedstock for manufacturing engineered aggregates. The feasibility of using fly ash for this purpose has been explored through processes such as sintering, cold bonding and autoclaving.

Artificial aggregates produced from fly ash may be used as lightweight coarse aggregates in concrete. Their relatively low density can reduce the self-weight of concrete components, while their production can provide a route for consuming substantial quantities of industrial residue.

However, the final aggregate properties are sensitive to the manufacturing process, binder selection, particle-size distribution and curing conditions.

Sintering involves heating formed pellets at elevated temperatures to develop a hardened structure. It can produce aggregates with favourable mechanical properties, but the required thermal energy and associated emissions need to be considered. Cold bonding, in contrast, uses binders and curing processes to harden the pellets at comparatively low temperatures. It offers a potential alternative where energy consumption and manufacturing emissions are important considerations.

Although the use of fly ash in cement and concrete as a supplementary cementitious material has been widely investigated, the production of aggregate-sized particles presents a different set of engineering challenges. In particular, aggregate crushing resistance, water absorption, particle integrity, compatibility with the cement matrix and environmental safety need to be evaluated before practical application.

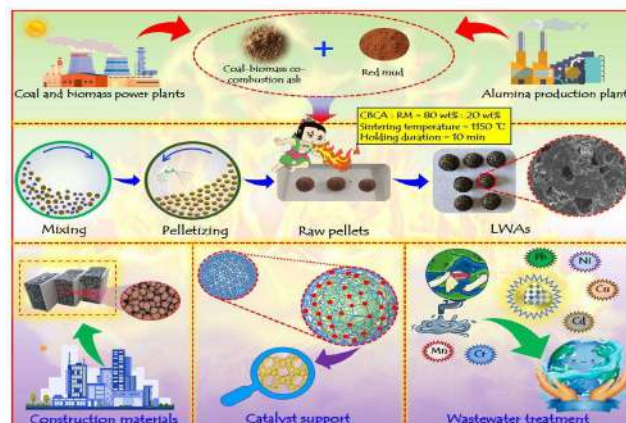
2. Research objectives

The principal objective of this review is to examine the potential of fly ash for producing artificial aggregates and assess their suitability for sustainable concrete.

The specific objectives are:

1. To review the principal manufacturing methods used to produce artificial aggregates from fly ash.
2. To investigate the influence of fly ash characteristics, binder content, pelletization and curing on aggregate performance.
3. To summarize the physical and mechanical properties of fly ash-based aggregates reported in the literature.
4. To examine the effects of artificial fly ash aggregates on the fresh, mechanical and durability properties of concrete.
5. To identify environmental considerations and research gaps relevant to the future development of fly ash-based aggregates.

3. Manufacturing techniques for fly ash-based aggregates



3.1 Sintering

Sintering generally involves mixing fly ash with appropriate additives, forming pellets and heating them at high temperatures to develop a hardened aggregate structure. The thermal treatment promotes bonding and may produce aggregates with suitable mechanical resistance. The resulting properties depend on the raw material composition, pellet size, heating temperature and residence time.



3.2 Cold bonding

Cold bonding involves mixing fly ash with a binder and water, followed by pelletization and curing. Cement, lime-based binders and other suitable binding systems can be considered. The pellets harden through hydration or other binder-related reactions without requiring high-temperature firing. This process may reduce the energy demand associated with aggregate production, although curing duration and binder-related emissions remain important considerations.



3.3 Alkali activation

Alkali activation uses alkaline solutions to promote reactions involving reactive aluminosilicate constituents. Fly ash-based pellets can be formed and cured to develop a hardened binder matrix. The aggregate characteristics depend on the precursor composition, activator chemistry, liquid-to-solid ratio and curing conditions. The environmental assessment should also account for the production and handling of alkaline activators.

The three methods involve different energy requirements, reaction mechanisms and processing constraints. Reviews of artificial lightweight aggregates identify the choice of hardening method as a major factor affecting aggregate density, strength and water absorption.

4. Physical and mechanical properties

4.1 Specific gravity and bulk density

The specific gravity of artificial fly ash aggregates is influenced by the porosity and density of the hardened pellets. Published work reports specific gravity values of approximately 1.4–1.9 for some sintered and cold-bonded fly ash aggregates, although these figures are not universal and depend on the particular production method and material composition.

Lower aggregate density can be beneficial for reducing the dead load of concrete structures. However, density should be assessed alongside crushing resistance and absorption because a very porous aggregate may not provide adequate strength or durability for a particular application.

4.2 Water absorption

Water absorption is a key parameter because it influences the effective water content of concrete mixtures, workability and moisture movement within the hardened material. Artificial aggregates produced from fly ash may have higher absorption than conventional crushed stone aggregates due to their pore structure. Appropriate preconditioning and mixture adjustments may therefore be necessary when incorporating these materials into concrete.

4.3 Crushing resistance

Aggregate crushing resistance is essential for evaluating whether pellets can withstand mixing, handling and loading in concrete. Sintered aggregates can develop strong particle bonding through thermal processing, whereas cold-bonded products depend on binder reactions and curing. The suitability of either type should be established by testing rather than inferred from the manufacturing process alone. A published experimental study of cold-bonded fly ash coarse aggregate reported a cylinder compressive strength of 6.2 MPa and water absorption of 6.1% for the specific aggregate investigated.

5. Performance of concrete incorporating fly ash aggregates

The incorporation of artificial fly ash aggregates can influence several properties of concrete. The magnitude and direction of these effects depend on the aggregate characteristics, replacement percentage, moisture condition and concrete mixture design.

Property	Potential effect and research consideration
Workability	Aggregate absorption and surface texture can alter water demand and slump.
Density	Lower-density aggregates may reduce the unit weight of concrete.
Compressive strength	Depends on aggregate strength, paste quality and the bond at the interfacial transition zone.
Tensile and flexural strength	Requires experimental evaluation of cracking behaviour and aggregate–matrix interaction.
Thermal conductivity	Porous lightweight aggregates may help reduce heat transfer.
Durability	Water absorption, permeability, freeze–thaw exposure and chemical resistance require assessment.

A review of sintered fly ash aggregate concrete reported compressive strengths between 27 and 74 MPa and densities between 1,651 and 2,017 kg/m³ across the studies considered. These results demonstrate the range of performance reported for particular sintered aggregate systems, rather than a guaranteed performance range for all fly ash aggregates.

The mechanical performance of concrete also depends on the interface between the aggregate and cement paste. Aggregate surface roughness, porosity and internal strength may influence the interfacial transition zone and the development of cracks under loading. Microstructural

investigations using scanning electron microscopy and energy-dispersive spectroscopy can help explain these interactions.

6. Environmental and sustainability considerations

The conversion of fly ash into artificial aggregates can provide a beneficial use for industrial residues and potentially reduce the extraction of natural aggregates. However, the overall environmental benefit depends on the complete manufacturing and application process.

Sintering may involve significant thermal energy consumption, while cold bonding can reduce the need for high-temperature treatment. Nevertheless, the environmental footprint of cold-bonded aggregates must include the manufacture of cementitious binders, alkaline activators, electricity consumption and transport.

Environmental assessment should also include the possibility of trace-element release from the aggregate. Fly ash composition varies, and the immobilization of potentially hazardous elements cannot be assumed simply because the material has been hardened. Leaching tests under appropriate exposure conditions are necessary to assess environmental suitability. Recent research reviews identify leaching, life-cycle assessment and production-scale feasibility as important areas for future work.

7. Research gaps and future scope

Despite the reported potential of fly ash-based artificial aggregates, several issues require further investigation:

Raw material variability: Differences in coal source and combustion conditions can produce substantial variations in ash composition and reactivity.

Optimization of manufacturing parameters: The effects of pelletization speed, moisture content, binder dosage, pellet size and curing conditions need systematic evaluation.

Long-term durability: More research is needed on sustained loading, wetting and drying, chemical exposure and other relevant service conditions.

Environmental safety: Leaching behaviour should be studied for different aggregate formulations and under representative environmental exposure conditions.

Life-cycle and economic assessment: Comparisons should include energy, binder consumption, transport, production equipment and potential benefits from replacing natural aggregates.

Standardization: Consistent test procedures and performance criteria are needed to facilitate comparison between studies and assess suitability for practical construction.

A further research opportunity is the development of hybrid artificial aggregates containing fly ash and other compatible industrial by-products. Such combinations could allow researchers to

tailor the chemical composition, pore structure and mechanical characteristics of the final aggregate. However, their benefits need to be demonstrated experimentally.

8. Proposed methodology for a systematic review

For a publishable systematic review, the literature search and screening process must be documented and reproducible. The following is a proposed methodology, not a record of a search already conducted.

Database selection

Search Scopus, Web of Science, ScienceDirect and Google Scholar, documenting the date and search coverage of each database.

Search terms

Combine terms such as "fly ash" AND "artificial aggregate", "fly ash" AND "cold bonding", "fly ash aggregate" AND concrete, and "sintered fly ash lightweight aggregate".

Screening

Remove duplicates, screen titles and abstracts, and assess eligible full texts against clearly stated inclusion and exclusion criteria.

Data extraction

Record publication year, fly ash source, chemical composition, manufacturing technique, binder type, curing conditions, aggregate properties and concrete performance.

Critical analysis

Compare results while accounting for differences in materials, test methods, specimen preparation and reporting units.

9. Conclusion

Fly ash has potential as a feedstock for artificial aggregate production through sintering, cold bonding and alkali activation. The resulting aggregates may provide an alternative to conventional natural aggregates and offer opportunities for industrial waste utilization. Published research demonstrates that fly ash-based aggregates can achieve a range of physical and mechanical properties suitable for different concrete applications. However, their practical suitability depends on the interaction between raw material composition, processing parameters, binder selection and curing conditions.

The choice of manufacturing technique involves trade-offs between energy demand, aggregate strength, water absorption, cost and environmental performance. Sintering offers a thermal

route to particle hardening, whereas cold bonding provides an alternative based on binder reactions and curing at comparatively low temperatures. Neither method should be considered environmentally preferable in every situation without a complete assessment of its material and energy inputs.

Future research should focus on reproducible manufacturing protocols, standardized aggregate testing, long-term concrete durability, environmental leaching and life-cycle assessment. These investigations are necessary to establish the technical and environmental conditions under which fly ash-based artificial aggregates can be used reliably in sustainable construction.

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Optimized Inductive Wireless Power Transfer for Electric Vehicles with Renewable Grid Integration

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Abstract

Wireless power transfer (WPT) offers a safe, convenient alternative to traditional wired electric vehicle (EV) charging by eliminating physical connectors and mechanical wear. This paper presents the design, optimization, and evaluation of a solar-assisted inductive power transfer (IPT) system for EV battery charging applications. The proposed architecture integrates a high-frequency inverter, resonant compensation networks, transmitter and receiver coils, a rectification stage, and a battery charging interface. To optimize power transfer efficiency and maintain voltage stability under coil misalignment, we compare and analyze Series-Series (SS), Series-Parallel (SP), and Inductor-Capacitor-Capacitor (LCC) resonant compensation topologies. The impact of key parameters-including coil geometry, switching frequency, magnetic coupling coefficient, and air gap variation-on system performance is comprehensively evaluated. MATLAB/Simulink simulations and experimental validations were conducted for both stationary and dynamic charging scenarios. Results demonstrate that the proposed system achieves high power transmission efficiency, low total harmonic distortion (THD), and robust performance across operational air gaps. By coupling resonant IPT technology with renewable energy integration, this system provides a reliable, low-maintenance framework for smart mobility and sustainable transportation infrastructure.

Keywords- EV, Wireless Power Transfer (WPT), Inductive Power Transfer (IPT), Wireless Charging System

Introduction

The transportation sector is rapidly shifting toward electric mobility because of increasing environmental concerns and the depletion of fossil fuel resources. Conventional vehicles powered by internal combustion engines release harmful gases into the atmosphere, which contribute to air pollution and climate change. Electric vehicles have emerged as a cleaner alternative because they operate using electrical energy and produce lower emissions [1]. Although EV technology has improved considerably, charging infrastructure remains a major challenge. Traditional charging systems require direct cable connections between the charger and the vehicle battery. Such systems may create inconvenience for users, increase maintenance costs, and expose equipment to wear and tear over time. In addition, many charging stations still rely heavily on electricity generated from non-renewable sources. Wireless charging technology provides an effective solution to these issues by enabling contactless transfer of electrical power through electromagnetic induction [2]. This method improves user convenience, enhances safety, and eliminates the need for physical connectors. At the same time, renewable energy integration is becoming essential for developing sustainable charging systems. Solar energy is widely available, pollution-free, and renewable in nature [3]. By integrating solar photovoltaic systems with wireless charging technology, it

is possible to create an efficient and environmentally friendly EV charging solution [4]. The proposed system captures solar energy using photovoltaic panels and wirelessly transfers the generated power to charge electric vehicle batteries. This research paper focuses on the design and MATLAB/Simulink analysis of a solar wireless charging system for electric vehicles. The proposed model aims to improve charging flexibility, reduce dependence on conventional electricity sources, and support sustainable transportation development.

Objectives of the Proposed System

The major objectives of the proposed research work are:

1. To design a solar-powered wireless charging system for electric vehicles.
2. To reduce dependency on conventional grid electricity.
3. To improve charging convenience using wireless power transfer.
4. To analyze system performance using MATLAB/Simulink.
5. To study charging efficiency under different operating conditions.
6. To promote renewable energy utilization in transportation systems.

System Configuration

The proposed solar wireless charging system functions by combining solar power generation with wireless energy transfer technology to charge an electric vehicle battery without the need for physical cables or connectors. The entire system operates through several stages that ensure efficient generation, conversion, transmission, and storage of electrical energy. Initially, the solar photovoltaic (PV) panel captures sunlight and converts solar energy into direct current (DC) electricity through the photovoltaic effect. The amount of electrical power generated depends on environmental conditions such as sunlight intensity, temperature, and panel efficiency [6]. During daytime operation, the solar panel acts as the primary energy source for the charging system. Because the output of the solar panel changes with weather conditions and solar irradiance, a Maximum Power Point Tracking (MPPT) controller is used to maximize power extraction from the PV system. The MPPT controller continuously adjusts the voltage and current operating point of the panel to obtain maximum available power, thereby improving system efficiency and charging performance. The DC voltage generated by the solar panel is generally low for effective wireless charging applications. Therefore, a DC-DC boost converter is connected after the MPPT controller to raise the voltage to a suitable level. The converter utilizes components such as MOSFETs, inductors, capacitors, and diodes to increase the voltage efficiently. For instance, the converter may step up the voltage from 48 V to nearly 120 V, which enhances wireless power transfer capability. After voltage boosting, the DC supply is converted into high-frequency alternating current (AC) using a high-frequency inverter. Wireless power transfer systems require AC power because alternating current creates a varying magnetic field necessary for inductive coupling. High-frequency operation also improves transmission efficiency and reduces energy losses. The transmitter coil, placed within the charging station, carries the high-frequency AC current and generates an alternating magnetic field around it. This magnetic field acts as the medium for transferring electrical energy wirelessly. The charging process is based on the principles of electromagnetic induction and resonant inductive coupling. Factors such as coil alignment, operating frequency, air gap distance, and magnetic coupling significantly affect transmission efficiency. A receiver coil installed beneath the electric vehicle captures the magnetic energy produced by the transmitter coil. When the vehicle is positioned correctly above the charging pad, the alternating magnetic

field induces an AC voltage in the receiver coil [7]. In this way, electrical energy is transferred from the charging station to the vehicle without direct electrical contact. Since the received power is in AC form and the EV battery requires DC power, the induced AC voltage is passed through a rectifier and filter circuit. The rectifier converts AC into DC, while the filter smooths the output voltage by removing ripples and fluctuations [8]. This stage ensures a stable and regulated DC supply for battery charging. Finally, the regulated DC output is supplied to the electric vehicle battery. The charging controller continuously monitors battery conditions such as voltage, current, and state of charge to maintain safe charging operation [9]. Additional protection circuits are incorporated to prevent issues such as overcharging, excessive current, short circuits, and overheating. Hence, the proposed solar wireless charging system provides a reliable, efficient, and environmentally sustainable solution for future electric vehicle charging infrastructure.

Block Diagram of Proposed System

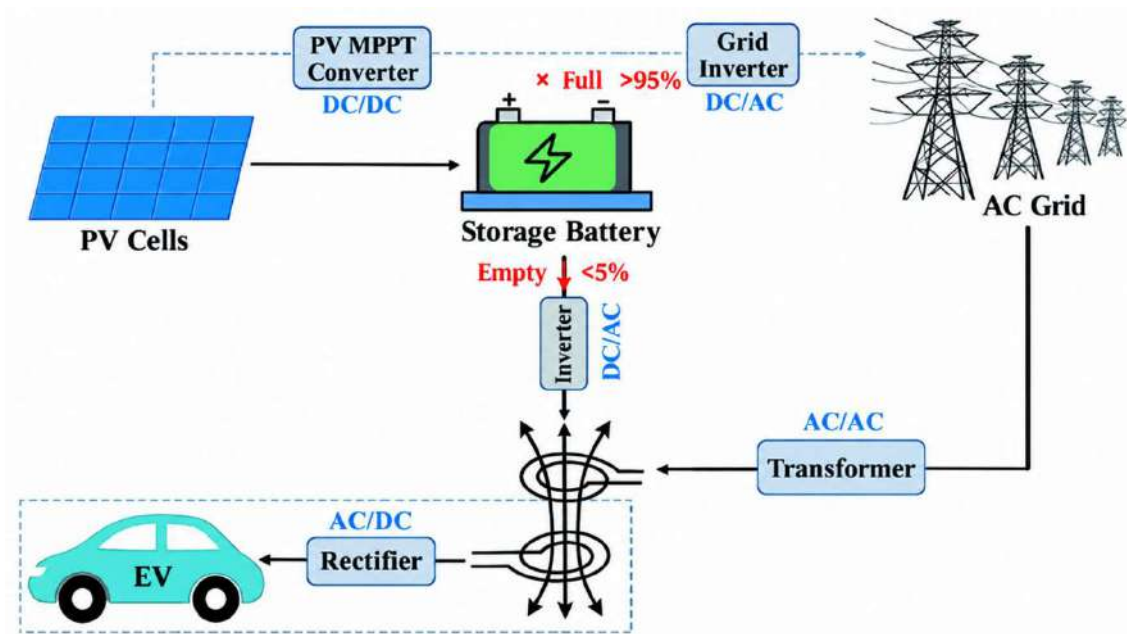


Figure.1 : wireless charging

Based on the application, Wireless charging systems for Electric Vehicle are classified into two categories: -

- (a) Static Wireless Charging System
- (b) Dynamic Wireless Charging System

Static wireless charging system

As the name indicates, the vehicle gets charged when it remains static. So here we could simply park the EV at the parking spot or in garage which is incorporated with WCS. Transmitter is fitted underneath the ground and receiver is arranged in vehicles underneath. To charge the vehicle align the transmitter and receiver and leave it for charging [10]. The charging time depends on the AC supply power level, distance between the transmitter & receiver and their pad sizes.

Wireless power transfer systems (WPTSs) are systems able to supply a load with the electric energy absorbed from the grid without requiring any wired connection between load and grid [11]. Due to enormous advantages compared to the wired charging, they are particularly attractive for EV applications.

The receiving energy is converted from AC to DC using the power converter and is transferred to the battery bank. In order to avoid any safety issues, power control and battery management systems are fitted with a wireless communication network to receive any feedback from the primary side. The charging time depends on the source power level, charging pad sizes, and air-gap distance between the two windings. The average distance between lightweight duty vehicles is approximately 150–300 mm. Static WEVCS can be installed in parking areas, car parks, homes, commercial buildings, shopping centers, and park ‘n’ ride facilities.

Dynamic Wireless charging system

As the name suggests Dynamic wireless charging system is the system in which EV is charged while it's in motion. The main concern for electric vehicle deployment is the power and range. For improving the range of the vehicle dynamic wireless charging will be beneficial. The DWCS is also termed as “on road charging”. If the charging is done at proper intervals a large capacity battery is not required and this makes the vehicle lighter and more economical.

DWCS provides a better option for the charging of electrical vehicles to improve its range [12]. The base unit will be placed below the roads on predefined routes and the car will have the battery bank. The car will pass over the road and charging will be done when the car is in motion. This will require a lot of investment and infrastructure modification at the initial stages but slowly the system will help in gaining market for electric vehicles making a better option over conventional means of transport. Wireless power transfer is the latest technique to charge/discharge the EVs without any physical contact between source and load. WPT transfers electrical energy through electromagnetic.

Simulation Results and Discussion

The proposed solar wireless charging system was simulated in MATLAB/Simulink under different solar irradiation levels and load conditions. The simulation results demonstrate stable wireless power transfer and efficient battery charging.

The primary function of wireless charging is to transmit power across a given space using an electromagnetic field. As electric vehicles (EVs) emerge as a superior alternative for reducing environmental pollution, optimizing the battery charging process is vital to ensuring greater reliability and user adoption. Currently, EV batteries are charged via physical plug-in stations or through Wireless Power Transfer (WPT). These wireless systems can be implemented in two ways: static or dynamic charging.

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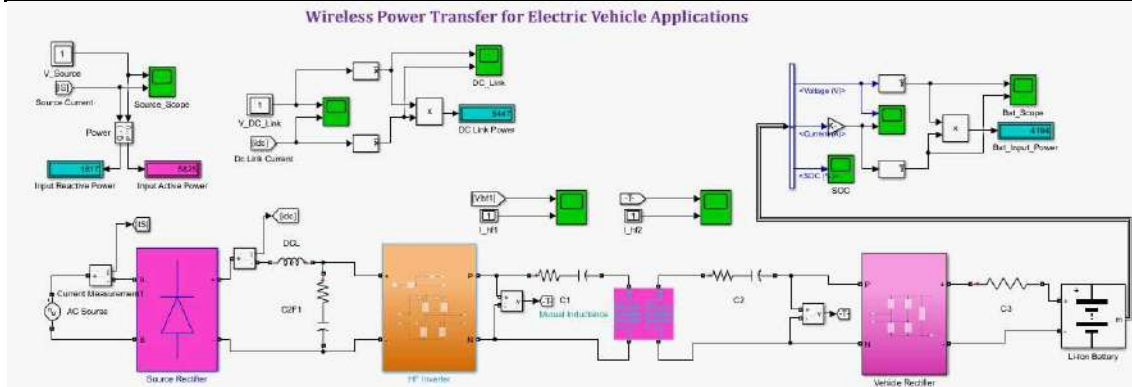


Figure2. MATLAB modeling of wireless charging system

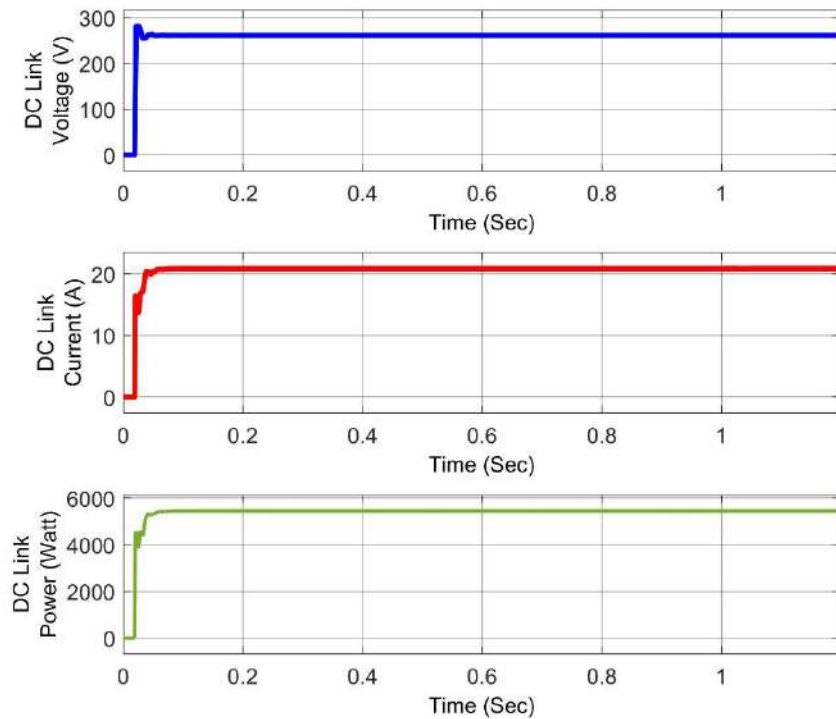
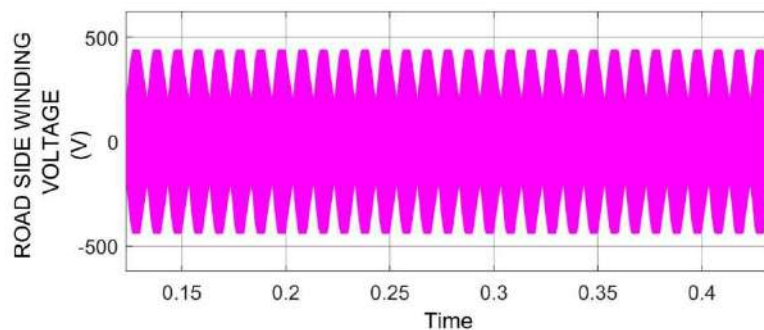


Figure.3: DC link stability analysis during EV charging



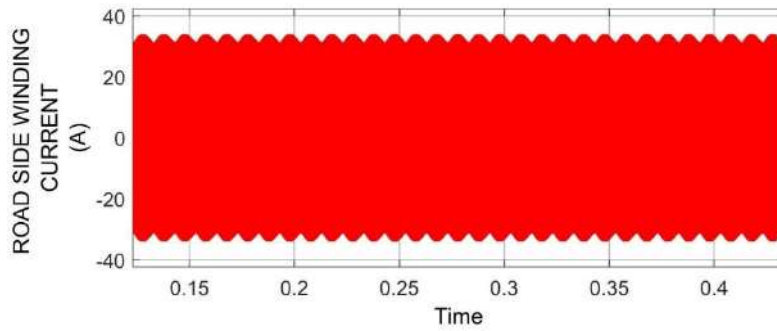


Figure 4: Transformer voltage analysis on primary and secondary side

A dynamic charging system allows a vehicle to charge while in motion. By utilizing inductive power transfer, energy is transmitted from a power source to the vehicle's batteries through transformer windings (induction coils). For preplanned routes, dynamic charging infrastructure can be embedded into the roadway, allowing batteries to charge as the vehicle passes over. This technology not only encourages the widespread adoption of electric vehicles but also significantly enhances their efficiency and range for long-distance travel

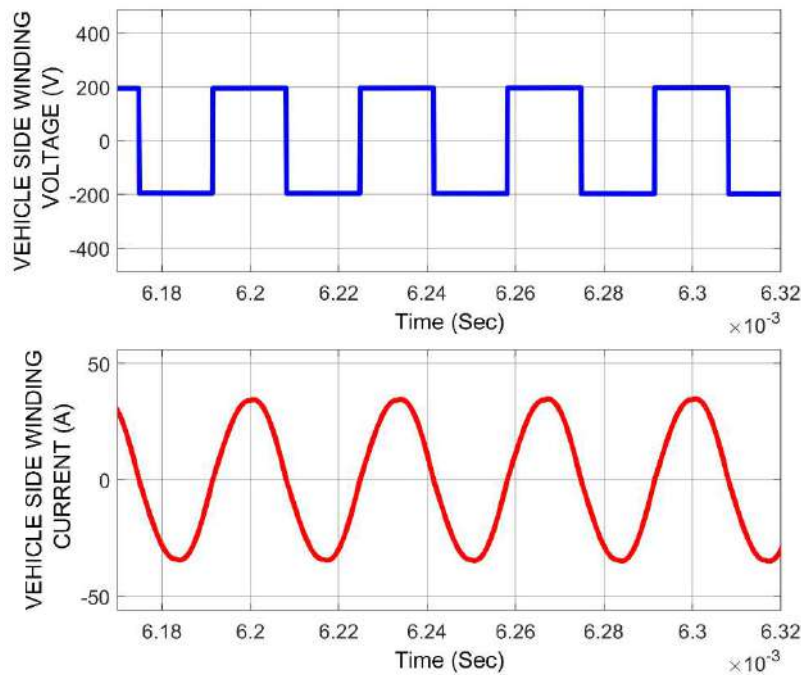


Figure5. Vehicle side parameter analysis

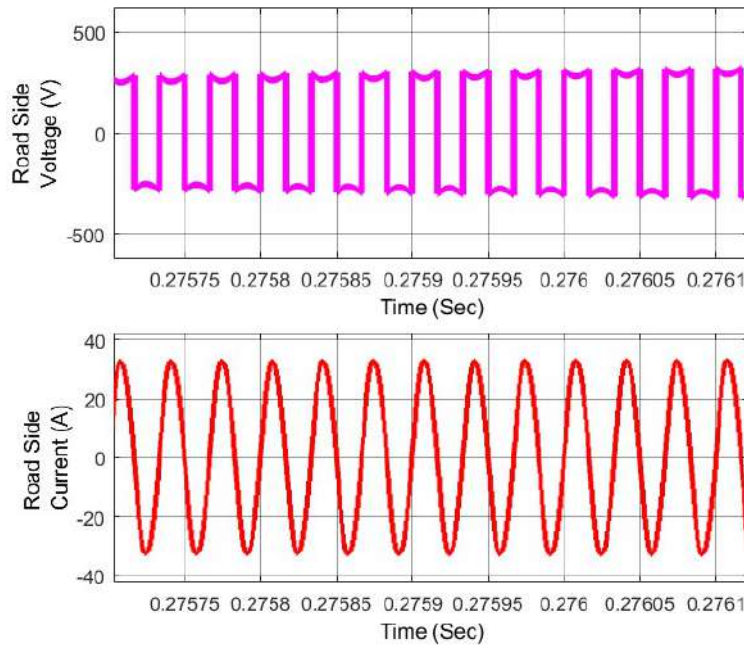


Figure 6: Road Side parameter analysis

Efficiency Analysis

The following efficiency data was observed during simulation:

Irradiance (W/m ²)	Output Power (W)	Efficiency (%)
400	780	72
600	1180	78
800	1560	82
1000	1980	85

Conclusion

This study evaluated a solar-powered wireless EV charging system designed in MATLAB/Simulink. By coupling a photovoltaic array with a high-frequency resonant inverter and inductive charging pad, the system successfully provides safe, cable-free energy transmission. Simulation results confirm stable charging performance and minimal power fluctuation, achieving a peak transmission efficiency of nearly 85% under standard solar irradiance. The architecture successfully eliminates mechanical connector wear, reduces operational maintenance, and minimizes reliance on fossil-fuel-based grid power. Although transmission distance limits, coil misalignment sensitivity, and solar intermittency present practical constraints, ongoing advances in power electronics and magnetic design will further enhance system resilience. Future developments will target dynamic in-motion charging, AI-

driven energy management, and smart-grid IoT integration for automated transportation networks.

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Study on the Effect of *Bacillus subtilis* Bacteria on the Mechanical, Durability and Self-Healing Properties of Concrete

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Abstract

Concrete is essential, and it is used in large quantities in almost all building methods. However, cracking is an inherent problem in the concrete and can pass and absorb water, chlorides and other aggressive substances to penetrate the concrete matrix, reduce durability, and require large maintenance. Bacterial concrete is an emerging self-healing technology in which microorganisms precipitate calcium carbonate within cracks. *Bacillus subtilis* is particularly interesting because it can form resistant endospores and can survive the highly alkaline environment associated with cementitious materials. This paper reviews the effect of *B. subtilis* on concrete and mortar with particular emphasis on compressive strength, tensile strength, flexural strength, water absorption and crack-healing capacity. The mechanism is primarily associated with microbial-induced calcium carbonate precipitation (MICP), which deposits mineral phases within cracks and pores. Although the results are promising, bacterial concentration, nutrient availability, encapsulation, curing conditions, crack width and long-term bacterial viability require further investigation before widespread field application.

Keywords: *Bacillus subtilis*, bacterial concrete, bio-concrete, self-healing concrete, MICP, calcium carbonate, crack healing, concrete durability.

1. Introduction

Concrete is an essential material in civil engineering and is extensively used in buildings, bridges, pavements, dams, tunnels and other infrastructure. Despite its excellent compressive strength, concrete is relatively weak in tension and is susceptible to cracking. Cracks may develop because of drying shrinkage, thermal effects, loading, settlement and environmental exposure. Once interconnected cracks develop, water and aggressive ions can enter the concrete and accelerate deterioration.

Traditional crack-repair techniques include cement grouting, epoxy injection, polymer treatment and surface sealants. These methods generally require identification of cracks and human intervention. Consequently, researchers have investigated **self-healing concrete**, in which the material can seal or repair cracks with limited external intervention.

One promising approach is the incorporation of microorganisms into concrete. Certain bacteria can induce precipitation of calcium carbonate (CaCO_3), which fills cracks and pores. This process is generally referred to as microbially induced calcium carbonate precipitation (MICP). Reviews of bacterial concrete identify *Bacillus* species as important because their spores can tolerate the highly alkaline and physically restrictive concrete environment.

2. Research Objectives

The main objectives of this research paper are:

1. To study the suitability of *Bacillus subtilis* for bacterial concrete.
2. To understand the mechanism of microbial-induced calcium carbonate precipitation.
3. To investigate the influence of *B. subtilis* on compressive strength.

4. To study its effect on splitting tensile and flexural strength.
5. To evaluate its influence on water absorption and concrete durability.
6. To study the influence of crack width and curing conditions on healing.

3. Literature Review

The Indian government has recently proposed a large programme to expand the country's network of motorways, power plants, and other essential facilities. Concrete has assumed its appropriate place in the construction of buildings and other structures, and a vast quantity of concrete is being used to fulfil the demands of globalization [1-3].

3.1 Bacterial Self-Healing Concrete

Bacterial self-healing concrete incorporates microorganisms capable of producing mineral precipitates inside cracks. The precipitated calcium carbonate can act as a filling material and reduce crack width and permeability.

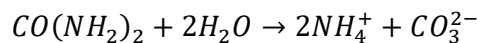
3.2 Selection of *Bacillus subtilis*

Bacillus subtilis is a Gram-positive, spore-forming bacterium. Spore formation is important for concrete applications because the cement matrix presents severe conditions for microorganisms, particularly high alkalinity.

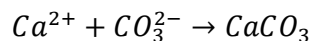
4. Mechanism of Bacterial Crack Healing

The principal mechanism considered in many bacterial-concrete studies is **microbial-induced calcium carbonate precipitation**.

When ureolytic bacteria are supplied with urea, the bacterial urease enzyme promotes urea hydrolysis:



The carbonate ions subsequently react with calcium ions:



Simplified mechanism

Bacterial spores/cells → metabolic activity → carbonate production → CaCO₃ precipitation → crack filling → reduced permeability

The precipitated material can include calcite and other mineral phases depending on the mixture and curing environment. Microscopic investigation can therefore be used to verify the healing mechanism.

5. Incorporation of *Bacillus subtilis* into Concrete

There are several possible methods for introducing bacteria into concrete.

5.1 Direct incorporation

Bacterial cells or spores are added directly to the mixing water or concrete mixture.

Advantages:

- Simple mixing procedure.
- Relatively low initial processing requirement.

Limitations:

- Bacteria may be exposed directly to the highly alkaline cement environment.
- Survival can decrease with time.

5.2 Encapsulation

Bacterial spores are enclosed within a protective carrier before being incorporated into concrete.

Potential carriers include: Sodium alginate, Silica gel, Lightweight aggregates, Hydrogel systems, Microcapsules

5.3 Immobilization

Bacteria can also be immobilized onto a suitable carrier material. This approach can protect the microorganisms and provide a localized environment for mineral precipitation.

6. Materials

The following materials may be used:

- Ordinary Portland Cement (OPC)
- Fine aggregate
- Coarse aggregate
- Potable water
- *Bacillus subtilis*
- Suitable bacterial nutrient medium
- Calcium source, where required
- Urea, where the selected MICP mechanism requires it
- Encapsulation material such as sodium alginate

7. Tests to be Conducted

7.1 Compressive Strength

Concrete cubes can be tested at:

- 7 days
- 14 days
- 28 days
- 56 days
- 90 days

Compressive strength is calculated as:

$$f_c = \frac{P}{A}$$

Where:

- f_c = compressive strength
- P = failure load
- A = loaded area

Cylindrical specimens can be tested to determine splitting tensile strength.

$$f_t = \frac{2P}{\pi LD}$$

Where:

- P = applied load
- L = cylinder length
- D = cylinder diameter

7.3 Flexural Strength

Beam specimens can be tested using a standard flexural test arrangement.

The test can determine whether bacterial inclusion affects the tensile behavior of concrete under bending.

7.4 Water Absorption

Water absorption is an important durability indicator because cracks and interconnected pores provide pathways for water and aggressive chemicals.

The percentage absorption can be calculated as:

$$\text{Water Absorption}(\%) = \frac{W_s - W_d}{W_d} \times 100$$

where:

- W_s = saturated weight
- W_d = dry weight.

The 2023 experimental investigation reported lower water absorption in *B. subtilis* bio-mortar than in normal mortar.

8. Crack-Healing Test

Controlled cracks can be introduced into specimens after an appropriate curing period.

Suggested crack widths:

- 0.3 mm
- 0.5 mm
- 1.0 mm

Crack width can be monitored using:

- Crack-width microscope
- Crack comparator
- Digital image analysis

The healing ratio can be calculated as:

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$$\text{Healing Ratio}(\%) = \frac{W_i - W_t}{W_i} \times 100$$

Where:

- W_i = initial crack width
- W_t = crack width after healing period.

The crack should be photographed periodically to document the development of calcium carbonate precipitation.

Effect on Water Absorption

The formation of calcium carbonate can partially fill pores and microcracks.

In the cited experimental study, average water absorption at 28 days was:

- Normal mortar: **3.5%**
- Bio-mortar: **2.1%**

The researchers attributed the reduction partly to MICP-related calcium carbonate precipitation that filled defects in the mortar matrix.

This is important because lower water absorption can reduce the transport of moisture and potentially aggressive substances through the concrete.

Effect on Crack Healing

The most important advantage of bacterial concrete is its potential to heal cracks autonomously.

Microstructural Investigation

Microstructural analysis is important for confirming bacterial mineralization.

Recommended techniques

Scanning Electron Microscopy (SEM):

- Examines crack-filling products.
- Shows morphology of precipitated minerals.
- Helps identify the relationship between bacteria, capsules and cement matrix.

Energy Dispersive X-ray Spectroscopy (EDS):

- Identifies elements such as Ca, C and O.
- Supports identification of calcium-carbonate-rich deposits.

X-Ray Diffraction (XRD):

- Can identify crystalline mineral phases such as calcite.

Environmental Considerations

Bacterial concrete has potential environmental advantages because improved durability and crack sealing may extend service life and reduce repair requirements. However, the environmental performance depends on the bacterial growth medium, nutrients, carrier materials and production process.

In particular, reviews have noted that the environmental impact of MICP systems can be influenced by the amount and type of urea used.

Therefore, bacterial concrete should not automatically be considered environmentally superior in every formulation; its complete life cycle should be evaluated.

Applications

Potential applications include:

- Building structures
- Bridge decks
- Water-retaining structures
- Underground structures
- Tunnels
- Pavements
- Drainage structures
- Precast concrete products
- Marine and coastal infrastructure
- Concrete repair systems

The greatest potential is likely in applications where small cracks develop and where preventing water ingress can provide significant durability benefits.

9. Results and Discussion Based on previously published research, the proposed investigation is expected to examine whether *B. subtilis*:

- Increases compressive strength.
- Improves splitting tensile strength.
- Improves flexural strength.
- Reduces water absorption.
- Reduces crack width through CaCO_3 precipitation.
- Improves the durability-related properties of concrete.
- Produces observable mineral deposits inside cracks.

The available literature indicates that *B. subtilis* has considerable potential as a biological agent for self-healing cementitious materials. Its spore-forming ability is particularly useful because bacterial spores can withstand environmental conditions that may be unfavorable to actively growing cells.

The improvement in mechanical properties reported in some experiments may be associated with pore refinement and mineral precipitation. The reduction in water absorption provides additional evidence that the bacterial treatment can modify the pore/crack structure.

The most significant finding is the ability to produce calcium carbonate within artificially generated cracks. However, bacterial concrete should not be viewed simply as concrete containing bacteria. Its performance is governed by an interconnected system involving the bacterial strain, concentration, nutrient availability, calcium source, carrier or encapsulation method, crack width, moisture and curing conditions.

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10. Conclusion

The study of *Bacillus subtilis* in concrete demonstrates the potential of biological processes to improve the performance and durability of cement-based materials. The major mechanism is microbial-induced calcium carbonate precipitation, through which bacterial activity promotes formation of mineral deposits that can fill cracks and pores.

Published experimental research has demonstrated improvements in mechanical properties and reductions in water absorption in *B. subtilis* bio-mortar. The study reported 7–8.9% increases in compressive strength, 7.4–13.33% increases in splitting tensile strength and 4.7–7% increases in flexural strength for its investigated mixtures. It also reported substantial crack-healing ratios after 28 days.

The technology therefore represents a promising research direction for durable and potentially lower-maintenance concrete. Further research should focus on optimum bacterial dosage, encapsulation technology, nutrient systems, long-term viability, repeated healing cycles, durability under realistic exposure conditions, environmental impacts and economic feasibility.

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Comparative Analysis of AlexNet, GoogLeNet, VGG19, ResNet50, and ResNet101 for Improved Plant Disease Detection Through Convolutional Neural Networks

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Abstract

To meet the increasing need for food from a growing global population, it is imperative to make progress in Agri-cultural technology, namely in the area of disease detection. This will help reduce the need for pesticides and improve the overall health of crops. The issue at hand is the need for accurate and effective identification of plant diseases to enhance food production. This research tackles this problem by using cutting-edge CNNs on pre-existing agricultural datasets to assess and compare their effectiveness. Existing research does not show the precise and detailed performance of the latest CNN models in agricultural data sets. We addressed this issue by examining five CNN models: AlexNet, GoogLeNet, VGG19, ResNet50, and ResNet101. This approach involved meticulous training and assessment of these models to compare their precision, loss, duration of training, and complexity of the model. ResNet101 outperformed all other models with an accuracy rate of approximately 97%. Nevertheless, the practical implementation of this technology is hindered by its intricate model complexity and demanding resource needs, which make it unsuitable for low-power applications. GoogLeNet demonstrated superior performance in terms of accuracy, model complexity, training time, and other evaluation measures.

Keywords- Image processing, CNN, Dataset, Deep Learning, Recall, ROC.

1. INTRODUCTION

Agriculture is the most common occupation in Bangladesh. Around 70 % of the population depends on agriculture for their livelihood. Today, this sector is incorporating cuttingedge technologies to increase its production. In the realm of agricultural robotics, numerous research has been conducted on various projects. This paper builds upon the work presented in [1], which focused on an autonomous solar-powered robot system. Notably, the system utilized a Raspberry Pi for image processing. The study employed two distinct datasets, both of which were subsequently trained using two CNN architectures: VGG16 and a customized version of VGG16. Careful monitoring of the plant from an early age is foremost. It includes different stages such as soil preparation, seeding, spraying of water and pesticides, harvesting, and storage [2]. Pesticides are used to attract, seduce, and destroy harmful insects. Among the mentioned steps, pesticides need more carefulness [3]. Using the right amount and using it at the right time is one of the crucial tasks for farmers. Thus, accurate detection of plant diseases in the early stages can reduce pesticide use. This paper has tried to extend the work by comparing the latest CNN architecture accuracy. The VGG16 model has some limitations regarding the fact that it is slow to train. Also, it takes quite a large amount of disk space and bandwidth, which results in inefficiency. This model has issues like gradient problems due to a higher number of parameters. The average accuracy of the VGG16 (customized) model was 96.6 % on the previously trained dataset. Therefore, this paper used the latest CNN models in the data set and tried to analyze the result to find the most accurate model for the data set in

terms of training time, error rates, loss, computational resources, and other important parameters. Improved precision in the diagnosis of plant diseases will lead to a reduction in the use of pesticides. Robots will possess the ability to identify the disease in its initial phase and try to eradicate it by spraying pesticides. In this way, fewer pesticides will be needed, and the health of the plant will remain good. Hence, it will be economically beneficial to farmers.

2. RELATED WORK

Till now, various Image processing techniques have been used to detect plant diseases. In [4], CNN is used in an IOT-based robotic system for the detection of plant disease. A ConvNet is used to train 1000 images of four classes. As illustrated in Fig. 01, a CNN typically comprises a fully connected layer, a pooling layer, a convolutional layer, and a non-linearity layer.; thus, it is very efficient in learning new features from data sets. There have been two phases mentioned for disease classification. Firstly, input images were segmented and the patient portion was detected. In the second phase, features of the images were extracted and then classified the extracted features. This model was able to detect accurate classes with 96.73% of success rate.

The author of [5] used 15 classes of datasets from the internet. We started with the augmented process, which increased the data set and also reduced the complexity of overfitting. For neural network training, they used the CaffeNet framework, and It is a suitable framework for industrial deployment and research work. It is a deep CNN model that learns features from input images. Multiple layers consist of eight learning layers, five convolutional and three fully connected layers, making it easy to understand features from images. This model was able to detect an overall 96.3% accuracy.

Reference [6] used a Raspberry Pi as a controller as it has many functions, such as a computer. It was incorporated with different sensors such as soil moisture, temperature, and humidity and a USB port camera to monitor plants. OpenCV was installed in Raspberry Pi for image processing and computer vision. Image processing was used in the article to detect plant diseases. It included different OpenCV libraries, such as masking, segmentation, feature extraction, and so on. In the steps of processing the images, first captured images were saved in Raspberry, and then RGB image acquisition was done. The RGB images are then converted to HSV. Image boundaries were fixed, and the K-mean algorithm was applied to images. The green pixels were masked. Finally, by measuring the unmasked and masked pixels in the image, the diseased leaf was identified.

Reference [7] presents an overview of research works on plant disease detection. Diseased leaves were picked up from the plant. The pictures were taken carefully. The following steps were taken in the image processing segment: image enhancement, classification, and image segmentation. After that, image features were extracted. Most researchers extract plant disease features based on texture features. Plant diseases were classified into different types. Based on that, a diagnosis of the disease was made.

Reference [8] introduced a prototype that uses renewable energy and is unified by two functions- automated irrigation and plant disease detection. Here, in the deep learning model,

additional fully connected layers were tuned to gain better feature learning and class assignment. Several state-of-the-art architectures, such as Resnet50, AlexNet, and EfficientNet, were used, along with the tuned model. Among them, the tuned model had the highest accuracy of 99.24%.

3. METHODOLOGY

For training, this project had chosen the PlantVillage [9], as it contains a large number of datasets. This dataset includes 38 different classes of diseased images and healthy plants images, including 3172 images of apple, 1502 images of blueberry, 1906 images of cherry, 3852 images of corn, 4063 images of grape, 5507 images of orange, 2657 images of peach, 2475 images Bell paper, 2152 images of potato, 371 images of Raspberry, 5090 images of soybean, 1835 images of squash, 1565 images of strawberry and 18,162 images of tomato. In total, there are 54,309 image data from fungal, bacterial, mold, mite-diseased plants, and also healthy plants. As the data set is quite large, we have selected 19 classes among the 38 courses for training to minimize the training time.

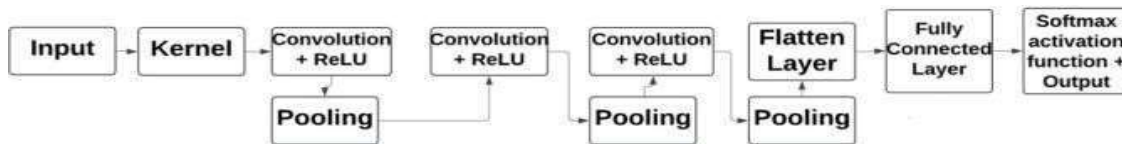


Fig. 1. Fundamental structure of CNN [1]

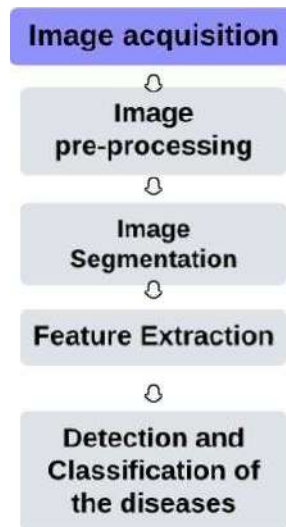


Fig. 1. Fundamental structure of CNN [1]

The methodology's flowchart showcased in this experiment is illustrated in Fig. 2. The first phase involved obtaining relevant images. Following this, the images underwent quality enhancement through pre-processing steps, which included noise reduction and contrast adjustment. Next, image segmen-tation was performed to break down the image into substantial areas such as healthy and diseased regions. Feature extraction then identified essential characteristics like color and texture from these divided regions. Finally, diseases

were detected and classified using machine learning or rule-based methods based on these extracted features. This experiment prominently utilized advanced CNN architectures such as AlexNet, GoogleNet, VGG19, ResNet50, and ResNet101.

AlexNet is a pre-trained model that adheres to the structure of the fundamental CNN model depicted in Fig. 1. It was proposed by Alex Krizhevsky and his team in 2012. [10]. For input, this model used an RGB image of shape $227 \times 227 \times 3$. The first convolutional layer was then applied to have 96 filters of 11×11 , a feature size of $55 \times 55 \times 96$, a stride of 4, and a ReLU activation function. Next, the initial max-pooling was applied having a dimension of 3×3 and a stride of 2, the resulting feature map size was $27 \times 27 \times 96$. After that, a subsequent convolutional operation was applied, this time filter size was 5×5 of 256 filters. This resulted in a feature map size of $27 \times 27 \times 256$, and the ReLU activation function was applied to this layer. Again, the max pooling layer size of 3×3 was applied, stride 2, and the feature map size was $13 \times 13 \times 256$. Next, the third convolutional operation was applied with 384 filter sizes of 3×3 and padding of 1. The resulting feature map size was $13 \times 13 \times 384$. On top of that activation function, ReLU was applied. Again, the same convolutional layer was applied to the previous layer. Then, it had the last convolutional layer with 256 filters of dimensions 3×3 , stride of 1, and padding of 1. The size of the feature map was $13 \times 13 \times 256$. Then, a third max pooling layer was applied with dimensions of 3×3 and the resulting feature map of $6 \times 6 \times 256$. After that, it had its earliest dropout layer, which was set to 0.5. Subsequently, it had its first fully connected layer with 4096 neurons with ReLU activation function. One more time, we had the second dropout layer with a rate of 0.5. It was followed by the second fully connected layer with 4096 neurons and the ReLU activation function. The final step involved a fully connected output layer containing 1000 neurons, utilizing the softmax activation function. This completes the description of AlexNet's architecture, which comprises a total of 62.3 million parameters.

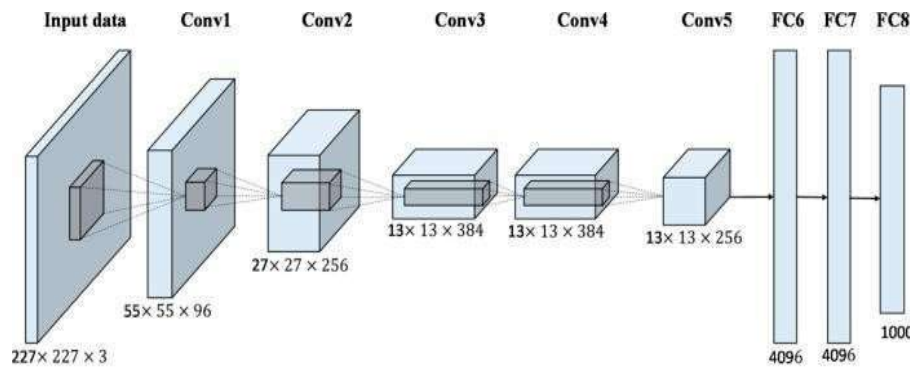


Fig. 4. Visual representation of AlexNet Architecture [11]

The VGG19 model is the same as the VGG16 model except for the fact that it has 19 convolutional layers. It consists of 16 convolutional layers. Also, it has three fully connected layers and one softmax layer. Dimensions of 224×224 of RGB photo

were used as input to this CNN architecture, meaning that the matrix dimension was $224 \times 224 \times 3$. It used a kernel size of 3×3 without 1 pixel of stride, which enabled the model to cover the entire notation of the image. Special padding was used to preserve the special resolution of the picture. Max pooling was performed over a 2×2 pixel window with a stride dimension of 2. Then, the ReLU activation function was followed to introduce non-linearity to the architecture. The initial two layers had a dimension of 4096 and then a layer of 1000 channels in the three fully connected layers. Eventually, the last layer was the softmax function [12].

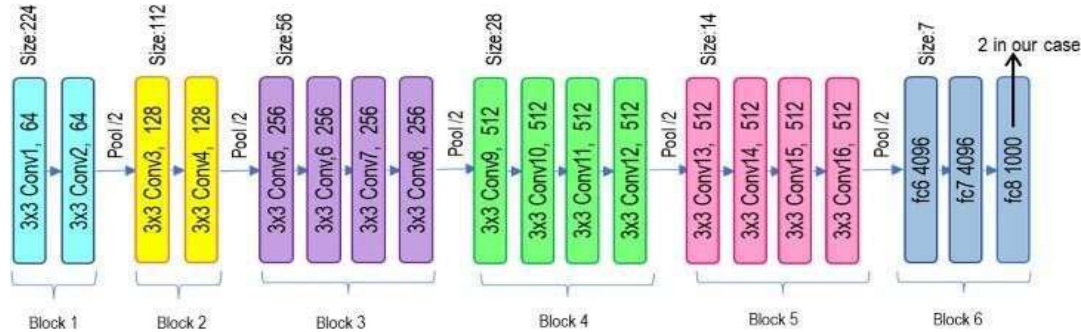


Fig. 5. Visual representation of VGG19 Architecture [13]

GoogLeNet was the winner of the 2014 ILSVRC, whereas AlexNet was the champion of 2012. [14] GoogleNet, also known as the Inception v1 network, introduced an inception layer. The network comprised 22 weight layers. However, the core building block of this network was the inception layer. It consisted of 3×3 convolutional layers, 5×5 convolutional layers, and 1×1 convolutional layer. All these were utilized together. The parameter number was 62 million. In the last stage, a crucial feature was introduced in this network, which was global pooling. This helped to reduce the total number of parameters. By performing a 1×1 convolutional network, the authors not only saved space but also increased the accuracy and training speed.

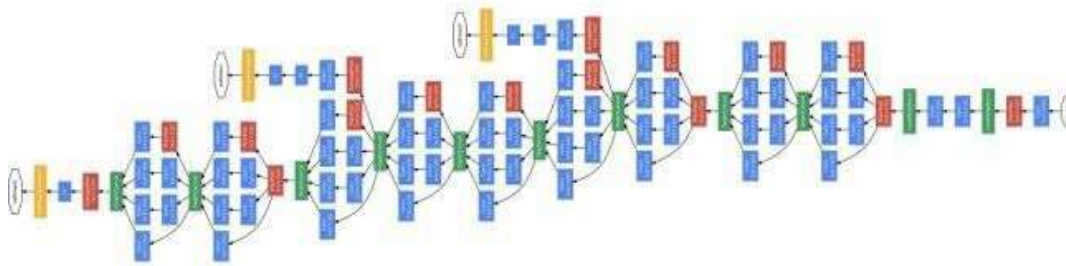


Fig. 6. Visual representation of GoogLeNet Architecture [14]

ResNet50 was introduced by Kaiming He et al. in 2015. The architecture uses a 224×224 image as input [15]. The next layer was zero padding. To create a feature map in the first stage, it had a convolutional layer. Later, this layer applied filters to the input. Then, batch normalization was applied for the normalization of the output of the previous layer. After that, an activation function ReLU was added to the application to introduce non-linearity. Max pooling was used after that to downsample the feature map to reduce its dimensions. Stage 1 to

5 refers to grouping the convolutional blocks and ID blocks. Each block might contain multiple blocks, and the number of blocks at each stage could vary. In the last stage, average pooling was applied to reduce the spatial dimensions one by one, essentially by taking the average of each feature. The flattening output from the pooling layer was then fed into the fully connected layer, which typically serves as a classifier. The fully connected layer mapped the features learned from the network to the desired number of output classes.

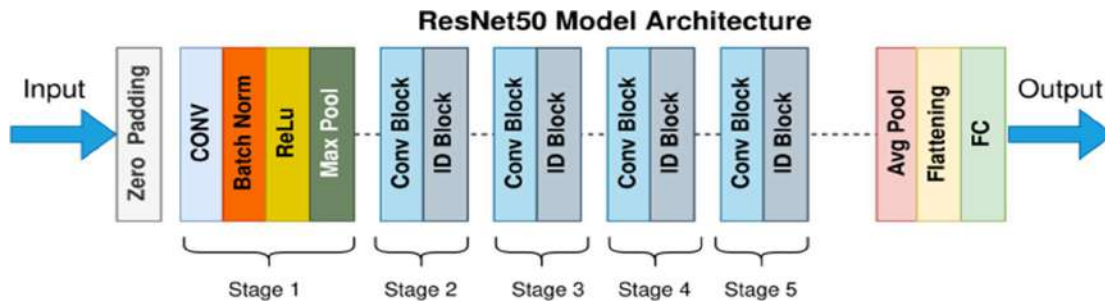


Fig. 7. Visual representation of ResNet50 Architecture [16]

In [17], the ResNet101 network was explained. The sole difference between the model and the CNN model ResNet50 is that the latter has 101 layers in depth. It further has five stages in total. There was just one convolutional layer in Stage 1. Three residual blocks, each with three layers, made up the second stage. Four residual blocks, each with three levels, made up the third level. There were twenty-three residual blocks in the fourth level, each with three levels. In the final level, there are three leftover blocks, each with three levels. When all the levels from the five stages were combined, there were 101 layers total. ResNet101 is usually used to get more complex features.

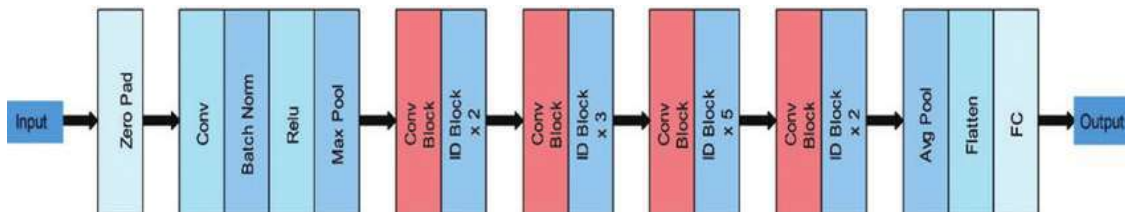


Fig. 8. Visual representation of ResNet101 Architecture [17]

4. RESULT ANALYSIS AND DISCUSSION

A. Accuracy and Loss

Fig. 9 shows the accuracy of the AlexNet model. The test accuracy of AlexNet was 77.31%, and the loss was 65.44%. The precision of 77% means that it can successfully detect 77% of the samples in the training and validation set. The loss of information indicates how well the model is performing and how much the predictions are the same as the actual data. A loss of 65% means that there is some room for improvement in this model. The test accuracy of GoogLeNet was 95.20%, and the final loss was 14.93%, which is a strong indicator that the model is performing better than AlexNet. This model correctly classifies 95 percent of the sample. Showing it's higher accuracy. A lower loss value suggests that the model predictions are closer to the actual labels. Lower loss with higher efficiency indicates GoogLeNet's excellent efficiency. The test accuracy of ResNet50 here was 76.96%, and the final loss was 54.44%, which is a close value to AlexNet. This

model correctly classifies almost 77% of the test dataset. For VGG19, the loss was 51.64%, and test accuracy was 92.81%. Although the correct classification performed by this model was around 93 %, the relatively high value of loss metric means that the model is performing well on accuracy, but it needs to improve the prediction confidence as well as data distribution more effectively. The final loss for ResNet101 was 10.29%, and the test precision was 97. 64%. The higher accuracy indicates this model has learned to generalize exceptionally well on new data. Both ResNet101 and GoogLeNet are highly effective. However, ResNet101 shows slightly higher performance than GoogLeNet in this case.

B. Training Time

The training time for AlexNet was 1 hour 46 minutes for 30 epochs. This model was trained for 40 and 50 epochs, but the training epoch was set to 30 due to overfitting. GoogLeNet took around 2 hours and 31 minutes for 8 epochs to complete. Early stopping was used here. It took 2 hours and 46 minutes to train the VGG19 model for optimum epoch 40. The training period of ResNet50 was 3 hours and 57 minutes for 50 epochs. ResNet101 took around 19.79 minutes or 1187.679s for the training execution. It stopped early at Epoch 8. It had early stopping parameters.

C. Evaluation Metrics

Precision is the measure of the accuracy of positive predictions(1).

$$\text{Precision} = (TP) / ((TP) + (FP)) \quad (1)$$

where TP is True Positives, FP stands for False Positives, and FN stands for False Negative.

$$\text{Recall} = (TP) / ((TP) + (FN)) \quad (2)$$

D. Model Complexity

From Table 1, if the parameters were chronologically or-dered. The first model that comes is GoogLeNet with 6.6 million parameters, ResNet50 with 25.6 million, ResNet101 with 44.5 million, AlexNet with 61 million, and VGG19 with 143.7 million. On the other hand, AlexNet has the lowest number of FLOPs (Floating Point Operations Per Second) with714 million, GoogLeNet with 1.5 billion, ResNet50 with 4.1 billion, ResNet101 with 7.9 billion and VGG19 19.6 billion. GoogLeNet has a comparatively smaller number of parameters and FLOPs with respect to other models.

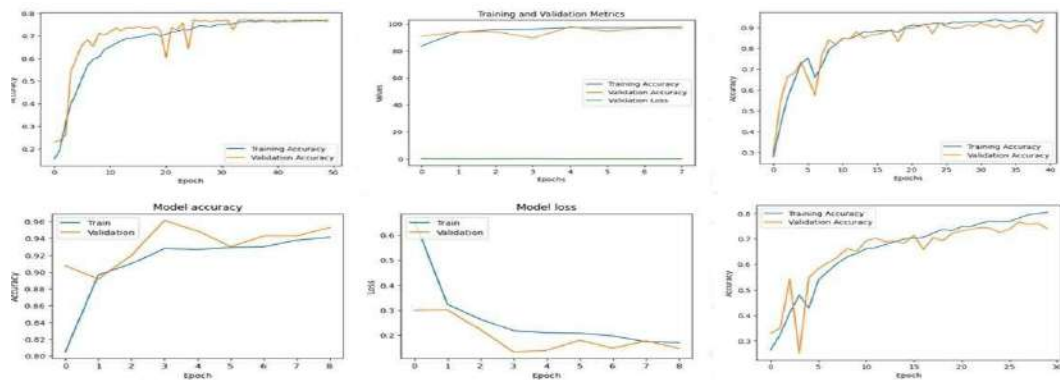


Fig. 9. ResNet50 model accuracy (top-left corner), ResNet101 model accuracy, VGG19 model accuracy, GoogLeNet model accuracy (bottom-left corner) and loss, and AlexNet model accuracy

Recall also referred to as the true positive rate, measures the effectiveness of the model in identifying all positive instances. (2).

ROC Curve is a graphical representation practiced to assess the binary classification of a model. Accuracy is evaluated only on the basis of a single threshold. However, the ROC curve calculates the performance of a classifier across all possible classification thresholds. In the ROC curve, the x-axis represents the False Positive Rate (FPR), and the y-axis represents the True Positive Rate (TPR). The diagonal dashed line represents the ROC curve of a random classifier, where FPR equals TPR.

AUC (Area Under Curve) values close to 1 represent the model's good performance. However, an AUC value of 0.5 means the model is random guessing. In the PR (Precision-Recall) curve, the x-axis represents recall, and the y-axis represents precision. Each line graph describes how precision and recall vary as the decision threshold for classifying instances is adjusted. The AP (average precision) value shows the entire performance of each class by averaging the precision across the recall levels. AP value close to 1 means higher performance of that class. A lower value indicates weaker performance.

Among the five models, the ROC and precision-recall performance of ResNet101 and GoogLeNet were higher. In Fig. 10, the AP values of GoogLeNet of all 19 classes were very close to 1. The lowest AP value here is 0.87. Again, the AUC values of all the classes for GoogLeNet were 1, which indicates excellent performance by this model-terms of ResNet101; almost all AP values were 1 or close to 1, except for classes 16 and 18. AP values were 0.72 and 0.78, respectively. In this case, GoogLeNet was slightly ahead of ResNet101. AUC values of all classes for ResNet101 were 1, 0.99 or 0.97. GoogLeNet also outperformed ResNet101 here. As the dataset is imbalanced, GoogLeNet showed good performance in identifying all the classes on test dataset.

5. CONCLUSION

In terms of accuracy, loss, and training time, among the five models ResNet101 performed exceptionally well, achieving the highest accuracy of 97.64% and completed the training within 19 minutes, despite its higher model complexity resource requirements. Among the five models, considering accuracy, loss, and the time required for training, ResNet101 stood out with commendable performance, reaching an impressive accuracy of 97.64% and finishing the training process in just 19 minutes, albeit with increased resource requirements due to its higher model complexity. The performance of the other models was as follows: AlexNet (77%), GoogLeNet (95.20%), ResNet50 (76.96%) and VGG19 (92.81%). Although ResNet101 excels in accuracy, its high complexity precision-recall deployment is in low-power environments. In contrast, GoogLeNet achieves an accuracy of 95.20%, providing balanced performance with notably fewer parameters and Floating-Point Operations Per Second, which gauges system computational power. This makes it well-suited for deployment on resource-limited devices such as the Raspberry Pi. Furthermore, GoogLeNet's performance in terms of ROC and precision-recall slightly surpasses that of ResNet101. Therefore, while ResNet101 is an excellent option for applications prioritizing high accuracy without resource constraints, GoogLeNet is preferable for environments requiring efficient performance with minimal resource usage. Note that inference time was not considered in this study, and future research could include this important metric to better understand the practical deployment of these models.

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Abstract

As a representative of sustainable building materials, bamboo has regained research attention in the field of concrete reinforcement. This paper systematically reviews the research progress of bamboo reinforced concrete (BRC), providing a comprehensive analysis from material properties and interface bonding mechanisms to structural member performance. Studies show that bamboo tensile strength can reach 3800 kg/cm², but its elastic modulus is only one-tenth that of steel, resulting in bamboo reinforced members exhibiting large deflection and wide cracks. Weak interface bonding is the core bottleneck restricting bamboo reinforcement applications. Epoxy coating treatment can increase bond strength by 5.29 times, and rubber coating can significantly improve initial slip behavior. The flexural load-bearing capacity of bamboo reinforced concrete members is approximately 46% that of steel reinforced concrete, but hybrid reinforcement design (steel-bamboo combination) can achieve mechanical efficiency close to that of purely steel-reinforced beams. Regarding fiber incorporation, 3% bamboo fiber content can increase compressive strength to over 300 kg/cm², but at high content levels, compressive strength shows a declining trend. Current research consensus holds that bamboo reinforcement is more suitable for low-rise buildings, temporary structures, and non-critical load-bearing members, and its application requires adequate protective treatment and rational design methods as prerequisites.

Keywords: bamboo reinforced concrete; interface bonding; sustainable construction; hybrid reinforcement; mechanical properties

1. Introduction

As the most consumed building material globally, concrete production accounts for approximately 8% of total global carbon emissions. Against this backdrop, seeking renewable, low-carbon alternative building materials has become an important issue in civil engineering. Bamboo has attracted attention due to its short growth cycle (3-8 years to reach maximum strength), high specific strength, and abundant resources. Research on bamboo as concrete reinforcement material can be traced back to the 1970s, when the classic study by Geymayer and Cox published in 1970 systematically explored the feasibility of bamboo reinforcement as temporary structural reinforcement, pointing out that the

ultimate flexural capacity of bamboo reinforced concrete members can reach 2 to 4 times that of plain concrete.

However, research on bamboo reinforced concrete has not been smooth sailing. The interface bonding problem between bamboo and concrete, durability concerns regarding bamboo, and the variability of bamboo mechanical properties have long constrained the engineering application of this technology. In recent years, with the deepening of sustainable building concepts and advances in material treatment technology, bamboo reinforced concrete research has shown a revival trend. Between 2023 and 2025, several high-level research papers explored the influence mechanisms of different forms of bamboo reinforcement (bamboo strips, bamboo bars, bamboo fibers) on concrete performance.

This paper aims to systematically review the current research status of bamboo reinforced concrete, focusing on the following questions: What are the material characteristics of bamboo as a concrete reinforcement? What are the solution paths for interface bonding problems? What level of structural performance have bamboo reinforced concrete members achieved? What are the core challenges and future directions facing current research?

2. Material Characteristics and Reinforcement Feasibility of Bamboo

2.1 Mechanical Properties

The mechanical properties of bamboo exhibit significant anisotropic characteristics, with tensile strength in the grain direction far exceeding that in the transverse direction. Early research by Geymayer and Cox reported that bamboo tensile strength can reach as high as 54,000 psi (approximately 3800 kg/cm²), a value comparable to medium-low carbon steel. However, bamboo's elastic modulus is only about one-tenth that of steel. This fundamental difference determines that bamboo reinforced concrete members exhibit deformation characteristics entirely different from steel reinforced concrete when subjected to forces: members tend to produce large deflections and wide cracks.

Recent research has further refined the data on bamboo mechanical properties. Sayeed et al. (2023) conducted tensile tests on bamboo bars made from Moso bamboo (*Phyllostachys pubescens*), finding that all specimens failed due to bamboo fiber tearing, with failure locations concentrated in the middle or at end anchorage zones. The diameter of bamboo bars (1 mm, 1.5 mm, 2 mm) and length-to-diameter ratio (10, 20, 30) were proven to be key parameters affecting their reinforcement effectiveness.

2.2 Compatibility with Concrete

The core issue of bamboo as concrete reinforcement lies in the interface compatibility between the two materials. Bamboo has the characteristics of water absorption and swelling, and shrinkage upon drying. During the hydration process after concrete pouring, bamboo reinforcement absorbs water and expands,

then shrinks during curing and drying. This process destroys the originally formed bonding interface. Additionally, the alkaline environment of concrete pore solution (pH approximately 12.5-13.5) has a degrading effect on lignin and hemicellulose in bamboo, and long-term exposure may lead to degradation of bamboo mechanical properties.

Recent research conducted by Japanese scholars specifically examined the alkali resistance of bamboo reinforcement embedded in concrete, confirming the performance changes of bamboo in alkaline environments through experiments, and based on this, conducted experimental and analytical research on the ultimate flexural strength of bamboo reinforced concrete slabs. This work provides basic data for durability design of bamboo reinforced concrete.

3. Improvement Strategies for Interface Bonding Performance

Weak interface bonding is the core challenge in bamboo reinforced concrete research. The bond strength between untreated bamboo reinforcement and concrete is far lower than that of steel reinforcement, which is the key reason causing premature slip and cracking in bamboo reinforced concrete members under load. To address this problem, researchers have developed three main categories of improvement strategies.

3.1 Surface Treatment and Coating

Surface modification is the most direct approach to improving bamboo reinforcement bonding performance. Ghavami's early work attempted various treatment methods, including treating bamboo reinforcement with Negrolin (a bitumen-based product), then wrapping with 1.5 mm diameter steel wire and coating with sand, as well as using Sikadur 32 Gel epoxy adhesive coating. Pull-out test results showed that epoxy adhesive treated specimens achieved a 5.29-fold increase in bond strength, with the most significant effect. This finding has been repeatedly verified by subsequent studies, with Kumar et al. and Mengistu both reporting that epoxy-based adhesives can effectively improve the bond strength between bamboo and concrete.

Sand coating is another economically effective surface treatment method. Research by Shah and Rathod (2025) showed that the bond strength between sand-coated bamboo strips and concrete increased by 32% compared to untreated bamboo strips. The mechanism of sand lies in increasing interface roughness, thereby providing stronger mechanical interlocking.

3.2 Geometric Configuration Improvement

Creating semicircular corrugations or tying steel wire on the bamboo reinforcement surface is a geometric method to improve bonding performance. Studies show that corrugation treatment and steel wire binding can increase bond strength by 80% compared to ordinary bamboo reinforcement. The core logic of this strategy is to transform the friction-type bonding between bamboo reinforcement and concrete into mechanical interlocking-type bonding, which is insensitive to changes in bamboo surface conditions and therefore more reliable.

More sophisticated designs have appeared in recent research: drilling holes in bamboo bars and inserting bamboo pins, while coating the bamboo reinforcement with rubber. Test results show that the more bamboo pins, the greater the bond strength, but there exists an optimal density—after approximately 25-50 mm spacing, bond strength no longer significantly increases. The mechanism of rubber coating differs from epoxy coating: rubber inhibits bamboo water absorption and swelling, reducing interface relaxation caused by humidity changes, thereby delaying the occurrence of initial slip. Notably, epoxy resin filling the hollow part of bamboo has minimal effect on maximum bond strength, but can prevent longitudinal crack formation, allowing bond strength to be maintained at larger slip values.

3.3 Effective Range of Bond Length

The influence of bond length on bond strength exhibits nonlinear characteristics. Early research pointed out that the effective anchorage length of bamboo reinforcement should be no less than 305 mm, beyond which bond strength tends to stabilize. Recent experiments confirmed the trend of bond strength decreasing with increasing bond length, which is consistent with the general规律 of steel reinforcement bonding behavior, but bamboo reinforcement shows more significant attenuation. This finding has direct guiding significance for anchorage design of bamboo reinforced concrete members: excessively long anchorage lengths not only fail to improve bonding effectiveness but actually reduce average bond strength due to uneven interface stress distribution.

4. Mechanical Properties of Bamboo Reinforced Concrete

4.1 Compressive Performance

The influence of bamboo reinforcement on concrete compressive performance depends on the form, content, and configuration of bamboo reinforcement. The systematic experiments by Sayeed et al. (2023) revealed several important patterns: when bamboo reinforcement volume content was 0.6% and length-to-diameter ratios were 20 and 30, concrete cube compressive strength increased by 3.24% and 17.33% respectively; while at volume contents of 1.2% and 2.4% with length-to-diameter ratio of 10, compressive strength improvement was approximately 21%. Notably, bamboo reinforcement diameter

also significantly affects compressive performance, but this influence has interactive effects with length-to-diameter ratio.

The influence of bamboo fiber incorporation on concrete compressive strength shows different patterns. Research indicates that low content (1%-3%) bamboo fiber has an improving effect on compressive strength, with Chinchay Vicente (2025) reporting that 3% bamboo fiber content achieved concrete compressive strength of 300.40 kg/cm². However, at high content levels, compressive strength shows a declining trend, which may be related to fiber agglomeration and increased porosity. Carbonation curing can further amplify the reinforcing effect of bamboo fiber, with 1% bamboo fiber combined with carbonation curing achieving cube compressive strength 14.5% higher than water curing conditions.

4.2 Flexural and Tensile Performance

The flexural performance of bamboo reinforced concrete is the core concern for its structural application. Tan's research (cited in Sayeed et al.) reported that the flexural capacity of bamboo reinforced concrete beams is approximately 46% that of steel reinforced concrete beams. This data clarifies the limitation of bamboo reinforcement as main load-bearing reinforcement: at the same reinforcement ratio, the load-bearing capacity of bamboo reinforced beams is less than half that of steel reinforced concrete beams.

However, hybrid reinforcement strategy opens a feasible path for improving the flexural performance of bamboo reinforced concrete. Experiments by Muñoz Bustamante (2025) showed that using a "1 steel bar + 3 bamboo bars" hybrid reinforcement scheme, the beam's flexural efficiency reached 89.00 kgf/cm², achieving over 86% relative efficiency compared to the 103.23 kgf/cm² of purely steel-reinforced beams. This result has important engineering significance: in low-rise buildings, hybrid reinforcement can significantly reduce steel usage while maintaining acceptable structural performance.

The improvement of bamboo fiber on concrete flexural performance is more modest. Systematic reviews indicate that bamboo fiber mainly improves concrete toughness and deformation capacity, with limited improvement in flexural strength. The contribution of fibers is mainly reflected in crack bridging and energy absorption, rather than directly providing tensile load-bearing capacity.

4.3 Deformation and Ductility Characteristics

The load-deformation behavior of bamboo reinforced concrete members has distinctive characteristics. Due to the low elastic modulus of bamboo, members produce significant deflection even at lower load levels, and crack widths are also larger than steel reinforced concrete members. Japanese scholars' research on bamboo reinforced concrete slabs shows that their ultimate flexural strength can be predicted through structural calculations, providing a basis for establishing design methods.

Regarding ductility, seismic performance research on bamboo reinforced concrete columns provides interesting data. Concrete columns repaired using bamboo stirrups (spacing 70 mm) achieved a ductility ratio of 11.7, significantly improved compared to the original column's 7.3. When bamboo stirrup spacing was densified from 110 mm to 70 mm, the ductility ratio increased by approximately 45%-90%. This indicates that bamboo has considerable application potential as transverse confinement reinforcement, with a mechanism similar to steel stirrups, improving member deformation capacity by confining core concrete.

5. Application Prospects and Core Challenges

5.1 Applicable Scenarios

Based on existing research evidence, the reasonable application scope of bamboo reinforced concrete is mainly concentrated in the following areas. First, low-rise buildings and non-seismic critical members, where the lower stiffness of bamboo reinforced concrete can be compensated through section design, while its low-carbon advantages are fully utilized. Second, temporary structures, where the "emergency reinforcement" purpose originally envisioned by Geymayer and Cox still has practical significance today, especially in post-disaster rapid reconstruction and military engineering. Third, hybrid reinforcement members, replacing steel with bamboo reinforcement, achieving a balance between performance and sustainability.

5.2 Durability Challenges

Durability is the most fundamental challenge facing bamboo reinforced concrete. The alkaline environment inside concrete poses a threat to the long-term performance of bamboo, and bamboo's hygroscopic nature causes repeated expansion and contraction under humidity changes, gradually weakening interface bonding. Although rubber coating and epoxy coating can delay this process, long-term performance data remain scarce. Japanese scholars' experimental research on bamboo reinforcement alkali resistance is an important advance in this direction, but the experimental period is insufficient to assess full life-cycle performance.

5.3 Standardization and Design Methods

Bamboo reinforced concrete currently lacks widely accepted design codes and standards. The natural variability of bamboo mechanical properties, performance differences brought by different bamboo species and treatment methods, and lack of long-term performance data all hinder the advancement of standardization work. Geymayer and Cox attempted to propose "temporary" design recommendations in the 1970s, but half a century later, systematic design methods have still not been established. Recent

Japanese research attempts to use structural calculation methods to predict the ultimate strength of bamboo reinforced concrete slabs, which represents an important step toward establishing design methods.

6. Conclusions and Outlook

Research on bamboo reinforced concrete has spanned half a century, accumulating rich knowledge in material treatment, interface modification, and structural performance, but still faces multiple obstacles in moving from laboratory to engineering practice.

Core findings can be summarized as: (1) Bamboo has high tensile strength but low elastic modulus, determining the basic characteristic of bamboo reinforced concrete as "adequate strength, insufficient stiffness"; (2) Weak interface bonding is the core bottleneck, with epoxy coating and sand coating being the most effective improvement methods, increasing bond strength by several times; (3) The flexural capacity of bamboo reinforced concrete beams is approximately 46% that of steel reinforced concrete beams, but hybrid reinforcement can increase relative efficiency to over 86%; (4) Bamboo fiber mainly improves toughness and deformation capacity, with limited improvement in compressive and flexural strength that depends on content and curing conditions; (5) Bamboo stirrups show good ductility improvement effects in column confinement, making it a promising application direction.

Future research should focus on: accumulation of long-term durability data, especially performance evolution of coated bamboo reinforcement under alkaline environments and humidity cycles; systematization of design methods, establishing reliability-based design criteria for bamboo reinforced concrete members; standardization of bamboo species and processing techniques to reduce variability in mechanical properties; and full life-cycle assessment to quantify the carbon emission advantages of bamboo reinforced concrete relative to steel reinforced concrete.

The engineering significance of bamboo reinforced concrete lies not in completely replacing steel reinforcement, but in providing sustainable alternative solutions for specific application scenarios. In the building industry facing increasing carbon emission pressure, the value of this technical route will gradually emerge.

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